

1. What is DataSet in ADO.NET?

- A `DataSet` is an in-memory representation of data in ADO.NET. It can hold multiple tables, relationships, and constraints. `DataSet` is often used to retrieve and store data from a database without requiring a constant connection.

2. What are SQL Connection Methods?

- The primary SQL connection methods include:
 - **SqlConnection:** Establishes a connection to a SQL Server database.
 - **Open():** Opens a database connection.
 - **Close():** Closes an open connection.
 - **Dispose():** Releases the resources associated with the connection.

3. How to Detect a Mobile Device in jQuery?

- You can detect if the user is on a mobile device using jQuery with the following code snippet:

```
javascript
Copy code
if (/Mobi|Android/i.test(navigator.userAgent)) {
    console.log("Mobile device detected");
} else {
    console.log("Not a mobile device");
}
```

4. What is the Purpose of clone() in DataSet Methods?

- The `clone()` method creates a copy of the `DataSet`'s structure, including all `DataTables`, schemas, and constraints, but does not copy the data itself. This is useful when you need the structure without the data.

5. What is JSON?

- JSON (JavaScript Object Notation) is a lightweight data-interchange format. It is easy for humans to read and write, and easy for machines to parse and generate. JSON is often used to transmit data between a server and a web application.

6. What is SOAP?

- SOAP (Simple Object Access Protocol) is a protocol for exchanging structured information in the implementation of web services. It uses XML as its message format and typically relies on HTTP or SMTP for message negotiation and transmission.

Q2. Long Questions

1. What is a Static Method in JavaScript? Write a Program in JavaScript to Calculate the Area and Circumference of a Circle.

- A static method is a method that belongs to the class itself rather than to instances of the class. Static methods are called on the class itself, not on an instance of the class.
- Example JavaScript program to calculate the area and circumference of a circle:

```
javascript
```

```

Copy code
class Circle {
    static calculateArea(radius) {
        return Math.PI * radius * radius;
    }

    static calculateCircumference(radius) {
        return 2 * Math.PI * radius;
    }
}

const radius = 5;
console.log("Area:", Circle.calculateArea(radius));
console.log("Circumference:",
Circle.calculateCircumference(radius));

```

2. What is C# Exception Handling? Write a C# Program to Check Whether a Number is Armstrong or Not.

- **Exception Handling** in C# is a process that handles runtime errors. It uses `try`, `catch`, `finally`, and `throw` keywords to manage exceptions. `try` contains the code that might throw an exception, `catch` handles the exception, `finally` executes code after try-catch regardless of success or failure, and `throw` is used to explicitly throw an exception.
- Example C# program to check if a number is an Armstrong number:

```

csharp
Copy code
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int number = int.Parse(Console.ReadLine());
        int sum = 0, temp, remainder;
        int digits = number.ToString().Length;
        temp = number;

        while (temp != 0)
        {
            remainder = temp % 10;
            sum += (int)Math.Pow(remainder, digits);
            temp /= 10;
        }

        if (sum == number)
            Console.WriteLine($"{number} is an Armstrong
number.");
        else
            Console.WriteLine($"{number} is not an Armstrong
number.");
    }
}

```

- In the above C# program, the user inputs a number, and the program checks if it's an Armstrong number by calculating the sum of each digit raised to the power of the number of digits and comparing it to the original number.

Other coding EXAMPLES

1. C# Program to Find the Factorial of a Number Using Recursion

```
csharp
Copy code
using System;

class Program
{
    static int Factorial(int n)
    {
        if (n <= 1) return 1;
        return n * Factorial(n - 1);
    }

    static void Main()
    {
        Console.Write("Enter a number: ");
        int number = int.Parse(Console.ReadLine());
        Console.WriteLine($"Factorial of {number} is {Factorial(number)}");
    }
}
```

2. C# Program to Find the Largest Element in an Array

```
csharp
Copy code
using System;

class Program
{
    static void Main()
    {
        int[] numbers = { 10, 24, 3, 56, 72, 12 };
        int largest = numbers[0];

        foreach (int num in numbers)
        {
            if (num > largest)
                largest = num;
        }

        Console.WriteLine($"Largest number is {largest}");
    }
}
```

3. C# Program to Check if a String is a Palindrome

```

csharp
Copy code
using System;

class Program
{
    static void Main()
    {
        Console.Write("Enter a string: ");
        string input = Console.ReadLine().Replace(" ", "").ToLower();
        string reverse = new string(input.Reverse().ToArray());

        if (input == reverse)
            Console.WriteLine("The string is a palindrome.");
        else
            Console.WriteLine("The string is not a palindrome.");
    }
}

```

4. C# Program to Implement a Simple Banking System with Deposit and Withdrawal Methods

```

csharp
Copy code
using System;

class BankAccount
{
    private decimal balance;

    public void Deposit(decimal amount)
    {
        balance += amount;
        Console.WriteLine($"Deposited: {amount:C}");
    }

    public void Withdraw(decimal amount)
    {
        if (amount > balance)
            Console.WriteLine("Insufficient funds");
        else
        {
            balance -= amount;
            Console.WriteLine($"Withdrew: {amount:C}");
        }
    }

    public void CheckBalance()
    {
        Console.WriteLine($"Current balance: {balance:C}");
    }
}

class Program
{
    static void Main()

```

```

    {
        BankAccount account = new BankAccount();
        account.Deposit(1000);
        account.Withdraw(500);
        account.CheckBalance();
    }
}

```

5. C# Program to Sort an Array of Strings Alphabetically

```

csharp
Copy code
using System;

class Program
{
    static void Main()
    {
        string[] names = { "John", "Alice", "Zara", "Mike" };
        Array.Sort(names);

        Console.WriteLine("Sorted names:");
        foreach (string name in names)
            Console.WriteLine(name);
    }
}

```

JavaScript Coding Solutions

1. JavaScript Program to Reverse a String

```

javascript
Copy code
function reverseString(str) {
    return str.split('').reverse().join('');
}

const input = "hello";
console.log(`Reversed string: ${reverseString(input)}`);

```

2. JavaScript Program to Find Prime Numbers Between Two Numbers

```

javascript
Copy code
function isPrime(num) {
    if (num <= 1) return false;
    for (let i = 2; i <= Math.sqrt(num); i++) {
        if (num % i === 0) return false;
    }
    return true;
}

function findPrimes(start, end) {

```

```

    const primes = [];
    for (let i = start; i <= end; i++) {
        if (isPrime(i)) primes.push(i);
    }
    return primes;
}

console.log(findPrimes(10, 30));

```

3. JavaScript Function to Check if an Array is Sorted in Ascending Order

```

javascript
Copy code
function isArraySorted(arr) {
    for (let i = 0; i < arr.length - 1; i++) {
        if (arr[i] > arr[i + 1]) return false;
    }
    return true;
}

console.log(isArraySorted([1, 2, 3, 4])); // true
console.log(isArraySorted([4, 3, 2, 1])); // false

```

4. JavaScript Program to Convert Temperatures Between Celsius and Fahrenheit

```

javascript
Copy code
function celsiusToFahrenheit(celsius) {
    return (celsius * 9/5) + 32;
}

function fahrenheitToCelsius(fahrenheit) {
    return (fahrenheit - 32) * 5/9;
}

console.log(celsiusToFahrenheit(25)); // 77
console.log(fahrenheitToCelsius(77)); // 25

```

5. JavaScript Program to Find the Second Largest Number in an Array

```

javascript
Copy code
function secondLargest(arr) {
    let first = -Infinity, second = -Infinity;
    for (const num of arr) {
        if (num > first) {
            second = first;
            first = num;
        } else if (num > second && num < first) {
            second = num;
        }
    }
    return second;
}

```

```
console.log(secondLargest([3, 5, 1, 8, 7])); // 7
```

Advanced Coding Solutions

1. C# Program to Read Data from a Database and Display it in a Console Application

```
csharp
Copy code
using System;
using System.Data.SqlClient;

class Program
{
    static void Main()
    {
        string connectionString = "your_connection_string";
        using (SqlConnection conn = new
SqlConnection(connectionString))
        {
            conn.Open();
            SqlCommand command = new SqlCommand("SELECT * FROM
YourTable", conn);
            SqlDataReader reader = command.ExecuteReader();

            while (reader.Read())
            {
                Console.WriteLine($"{reader["Column1"]},
{reader["Column2"]}");
            }
        }
    }
}
```

2. JavaScript Calculator with Basic Operations

```
html
Copy code
<!DOCTYPE html>
<html>
<body>
    <input type="text" id="result" readonly>
    <button onclick="calculate()">=</button>
    <button onclick="clearDisplay()">C</button>
    <script>
        function calculate() {
            const result = document.getElementById('result');
            result.value = eval(result.value);
        }
        function clearDisplay() {
            document.getElementById('result').value = '';
        }
    </script>
</body>
```

```
</html>
```

3. C# Program to Handle Exceptions and Log Errors to a Text File

```
csharp
Copy code
using System;
using System.IO;

class Program
{
    static void Main()
    {
        try
        {
            int num = int.Parse("ABC");
        }
        catch (Exception ex)
        {
            File.WriteAllText("log.txt", ex.ToString());
        }
    }
}
```

4. JavaScript To-Do List with Add, Remove, and Mark Complete Features

```
html
Copy code
<ul id="todoList"></ul>
<input type="text" id="newTodo">
<button onclick="addTodo()">Add</button>

<script>
    function addTodo() {
        const list = document.getElementById("todoList");
        const todo = document.getElementById("newTodo").value;
        const listItem = document.createElement("li");
        listItem.textContent = todo;
        listItem.onclick = () =>
        listItem.classList.toggle("completed");
        list.appendChild(listItem);
    }
</script>
```

5. C# Console Application for a Quiz Game

```
csharp
Copy code
using System;

class Program
{
    static void Main()
    {
        int score = 0;
    }
}
```



```
        string[] questions = { "What is 2 + 2?", "What is the capital  
of France?" };  
        string[] answers = { "4", "Paris" };  
  
        for (int i = 0; i < questions.Length; i++)  
        {  
            Console.WriteLine(questions[i]);  
            string answer = Console.ReadLine();  
            if (answer.Equals(answers[i],  
StringComparison.OrdinalIgnoreCase))  
                score++;  
        }  
  
        Console.WriteLine($"Your score: {score}");  
    }  
}
```